Project 1 Cumulative Practical Assignment

- **Due** 30 Jan by 14:40

- **Points** 5

- **Submitting** a text entry box

- **Available** until 30 Jan at 16:00

This assignment was locked 30 Jan at 16:00.

Consider our Assignment

P1. Assignment 1.3 Admino Gradescope

CPA 1 Scenario

In this assignment you will be connecting to a NEW individual linux system hosted in the cloud. Each of you has assigned an individual port for connection (HOSTNAME and PORT have changed).

To connect you will be using SSH

- **user: student**
- **pass:15Champions**

Here is an example of the connection (**remember that you need to use a new own port to connect**):

$ssh student@cy5001CPA1.mscy.org -p 240XX

For this Project 1 cumulative Practical Assessment, you need to complete task7 in 30 minutes. You need to do it using your individual "user#" that you used for 1.3. Remember, you will be connecting to a fresh copy of your system.

7. Authentication based on public key

- We have stored a private key file "user" into the student home directory of your individual remote system. For this task your username will be unique.

- **Bring that private key from your remote system to your laptop**, and disconnect from the "student" connection (student@cy5001cpa1.mscy.org -p 240XX).
- Once at your laptop with the private key, use it to connect with **SSH to the following server cy5001cpa1.mscy.org at port 24100.**

Once you are in the remote system, you need to find a script named cookie and execute it providing your NUID as argument

```
>$ cookie 00123456789
00123456789fefb4f4847337a1bc28f0bf5e2bbad1ce4c14756
```

Once you obtain the cookie, you just need to include it as your answer for this assignment in canvas.

You will have **30 minutes** to complete the assignment. Your time will be tracked based on your connection your individual system (student@cy5001cpa1.mscy.org -p 240XX).

⚠️ **Important:** If you exceed the 30-minute limit, **1 percentage point will be deducted for each additional minute** you remain connected to the system (it does not matter if you are not performing any operations on the system.

Project 2 Cumulative Practical Assignment

- **Due** 20 Feb by 14:50

- **Points** 5

- **Submitting** a file upload

- **Available** 20 Feb at 14:00 - 20 Feb at 15:30 1 hour and 30 minutes

This assignment was locked 20 Feb at 15:30.

For this Practical assessment you are requested to complete a modified versions of tasks included in **Assignment 2.2 OpenSSL and digital signature** Parts 2 and 3.

Task 1. Decrypt a Symmetrically encrypted message intended to you

- In this part of the assignment you will be decrypting a symmetrically ciphered message (**AES-128** with CTR encryption mode and Salt added) that is stored in the shared folder /home/sharedfolder/cpa2 of the master server. You need to access to the master server and transfer (SCP) the stored message file, named userid.cpa2.enc.
  --> Remember to connect to the master server with the credentials provided by email and at cy5001.mscy.org -p 10010

- The file has been encrypted using the password you use to connect to the master server.

- Decrypt the file and store the decrypted message into a file named userid.cpa2.enc.flag

Task 2. Digitally sign the flag file you obtain as a result of the task 1 of this CPA

- Generate the digital signature of the flag file userid.cpa2.enc.flag you obtain in the previous task. Create the signature using **sha-512** digest algorithm with the private key that you generated during your assignment 2.2 Part 1 (userid.priv)**.**

- As a result of this task you need to obtain the userid.cpa2.enc.flag.sig file.

Deliverables for this CPA:

Make sure you submit in Canvas all the following files following the naming requirements:

1) userid.cpa2.enc.flag, 2) userid.cpa2.enc.flag.sig , 3) userid.pub

You have **40 minutes** to complete these tasks.

Project 3 Cumulative Practical Assignment

- **Due** 27 Mar by 14:40

- **Points** 5

- **Available** 27 Mar at 14:00 - 27 Mar at 15:00 1 hour

This assignment was locked 27 Mar at 15:00.

Project 3 Cumulative Practical Assignment

You will be working on task related with the P3. Assignment 3.1 Contapache. Secure Configuration of Web Server, However, you will use a different port and a fresh container prepared specifically for this CPA..

You will be connecting via SSH to **cy5001cpa3.mscy.org** and you can find your user and assigned ports in the table provided **HERE**.

ssh <your-userid>@cy5001cpa3.mscy.org -p <your-SSH-port>

- Use your individual system username and credentials (you received them by email in January and that you've used for several assignments.)

For this Practical Assessment you are required to configure HTTPS on your remote Apache CPA system.

- During Assignment 3.1 you created a self-signed certificate with the subject C=US, ST=MA, L=NEU, O=MSCY, CN=<your-userid>.cy5001.com and configured an Apache HTTPS server in your own system. We want you to use the same certificate (certificate and private key) in the new sistem prepared for you in this CPA.

Your system already has the following:

- Apache is running and serving HTTP on port 80, mapped to your assigned HTTP port on the host
- SSL configuration files (httpd.conf and httpd-ssl.conf). Verify paths, as they may differ from Assignment 3.1

- Port 443 is already mapped to your assigned HTTPS port on the host.
- Your individual user has sudo privileges in your individual CPA system.

TASKS you need to complete during this CPA:

1. Transfer your certificate and private key from Assignment 3.1 into the container (use scp or copy/paste)
2. Place the files in the correct locations as expected by the Apache configuration
3. Update the Apache configuration files as needed, ensuring that the appropriate modules and include directives are enabled
4. Restart Apache so the changes take effect (Apache binaries are located at: /usr/local/apache2/bin/)

Once you believe your configuration is correct, you can check it inside your container with curl or with your laptop browser connecting to https://cy5001cpa3.mscy.org and your individual HTTPS port.

You have 40 minutes to complete this task. The assessment ends at 2:40 PM (EST).

Project 4 Cumulative Practical Assignment

- **Due** 17 Apr by 14:45

- **Points** 5

- **Submitting** an external tool

- **Available** until 17 Apr at 15:40

This assignment was locked 17 Apr at 15:40.

Consider our Project 4 Assignment:

Project 4. Assignment 4.1 Zip Cracker in Python

For this Practical assessment you are requested to crack a NEW zipped file using your zipcracker program created for project 4.1.

Your individual NEW zipped-password-protected-file **in your home directory of the master server "/home/<username>/cpa4".**
> Remember that Master Server connection goes over port 10010
(ssh username@cy5001.mscy.org -p 10010)

```
@cy5001:/home/<username>/cpa4$ ls -la
total 8
drwxr-xr-x 2 root    root 4096 Apr  2 10:37 .
drwxr-x--- 4 <username> ta   4096 Apr  2 10:37 ..
-rw-r--r-- 1 root    root  312 Apr  2 10:37 <username>.1.zip
```

This time the dictionary file used to randomly select the password used to encrypt your zip file has been selected from a different dictionary. You need to create the dictionary file (like you did in *Task 4: Tailor a wordlist*).
The wordlist included in this new dictionary is described as:

- A fixed first name chosen from exactly one of: john, rick, mia, or arwa
- An optional hyphen (-) after the first name.

- A three-letter month abbreviation, one of: jan, feb, mar, apr, may, jun, jul, aug, sep, oct, nov, dec
- A two-digit day from 00 through 31
- A four-digit year between 2000 and 2026 inclusive

Here you have an example of a few passwords included in the dictionary.

arwaoct102013
rickmay052012
arwaapr272011
miajan282015
arwadec312012
miadec312001
john-feb152000
rick-mar112015
johnjun142009
arwa-nov312003
rickaug272017
johnsep292022
arwa-feb112023
miaoct302000
arwaoct222023
arwa-sep142006
rick-nov152015
johnmay132024
mia-aug302004
john-oct112016
mia-apr262017
arwa-aug012008
rickapr192014
rick-sep262020
arwa-apr202013
john-oct282018
arwa-dec022004
rickmar142011
john-dec192024
mia-feb202012

Once you generate the dictionary file according to the previous specifications, you can use your zipcracker program to crack the file contained into /home/<username>/cpa4. Once cracked, extract the flag from the zip archive and submit that flag included into a text file named <username>.cpa4flag.txt through Gradescope.

>> If you are not able to crack the flag, include the commands and/or code you have tried into a text file **<username>.cpa4flag.txt** and also submit it on Gradescope. So we can assess what you have tried.

Important: The deadline to submit this text file into grade scope is 40 minutes.